Lucas Rivero Iribarne

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Experience

Bunge Global SA **Rotterdam, The Netherlands**
Data Engineer *2025 – Present*
Desk: Research IT. Transitioned from Trainee to full-time Data Engineer. I develop ELT pipelines focused on geodata (Sentinel 1 & 2) and global weather data (ERA5, IFS ECMWF).
- Write and maintain Python ELT data pipelines to fetch business-relevant data.
- Dockerize Python pipelines and deploy them to Google Cloud Platform (GCP).
- Model warehouse data (BigQuery and Snowflake) using dbt to populate dashboards.
- Orchestrate and schedule pipelines using Apache Airflow.
- Build geospatial pipelines: Developed a scalable processing pipeline for Sentinel 1 and 2 data in Docker containers on AWS.
- Build processing pipelines to fetch ERA5 data and aggregate it using business-provided geometries.

Viterra Limited / Bunge Global SA **Rotterdam, The Netherlands**
Data Engineer Trainee *2024 – 2025*
Desk: Research IT. Completed a comprehensive traineeship program focusing on the fundamentals of Data Engineering, including Python, ELT, data modeling, SQL, and cloud computing.
- Applied Agile fundamentals in a team environment.
- utilized Python and Git for version control and software development.
- Developed transformations using Snowflake SQL and dbt.
- Created data visualizations and dashboards in Tableau.

Viterra Limited **Rotterdam, The Netherlands**
Crop Growth Modeling Intern *2024*
Thesis: "From pixels to information: Unveiling the prowess of crop growth models assisted by remote sensing and data assimilation techniques."
Desk: Research.
- Established a Crop Growth Model (CGM) pipeline (Python and Docker) using WOFOST for biomass and yield forecasting of Canola in Canada at the rural municipality level.
- Assimilated MODIS Leaf Area Index (LAI) using the Ensemble Kalman Filter to update the observed crop state.
- Updated and maintained a Postgres/PostGIS database with model inputs (weather, soil, management, regions, remote sensing data) and outputs.
- Calibrated Growing Degree Days (GDD) estimates to account for local cultivars.

eLEAF **Wageningen, The Netherlands**
Cloud Mask Re-labeler, Part-time *2023 – 2024*
- Re-labeled cloud mask product outputs using deep learning techniques to re-classify pixels, improving the overall accuracy of the model.

FONDECYT regular 1212071 **Santiago, Chile**
GIS Data Analyst *2021 – 2022*
"The catchment memory: understanding how hydrological extremes are modulated by antecedent soil moisture conditions in a warmer climate."
PI: PhD. Mauricio Zambrano Bigiarini
- Developed a methodology to assimilate LAI data from remote sensing into the Soil Water Assessment Tool (SWAT+) to improve streamflow/evapotranspiration simulations and reduce model equifinality.
- Calibrated the SWAT+ model to simulate water flow and hydrological processes at the catchment level.
- Led, planned, and coordinated field campaigns to retrieve data from rain gauges and soil moisture sensors in four Chilean watersheds.
- Wrote scripts to process field data into actionable information and visualizations.
- Developed web scraping scripts to gather meteorological time series data from national systems.
- Automated the downloading of evapotranspiration (PML) and soil moisture (SMAP and SMOS) products for all of Chile.

FONDECYT regular 1210932 **Santiago, Chile**
Research Assistant *2021*
"Improving forest water yield and productivity quantification at the catchment scale by mapping root depth and eco-physiological thresholds with remote sensing and water transfer modeling"
PI: PhD. MSc. Mauricio Galleguillos Torres
- Wrote scripts to pre- and post-process sap flow data recovered from in-situ sensors.
- Trained a junior research assistant to ensure project continuity.

University of Chile **Santiago, Chile**

Assistant – PCI CONICYT Chile, NSFC190018 *2021*

"Management of global change impacts on hydrological extremes by coupling remote sensing data and an interdisciplinary modelling approach"

PI: PhD. Mauricio Zambrano Bigiarini

- Planned, coordinated, and led field campaigns to install rain gauges in four Chilean watersheds.
- Performed rain gauge calibration.
- Selected optimal installation sites for rain gauges based on project objectives.

FONDECYT regular 1171560 **Santiago, Chile**

Research Assistant *2019 – 2021*

"Assessing spatio-temporal impacts of global change on water and biomass production processes at catchment scale: a synergistic approach based on remote sensing and coupled hydrological models to improve sustainable management of forest ecosystems"

PI: PhD. MSc. Mauricio Galleguillos Torres

- Led field campaigns to measure ecosystem variables (soil physics, soil moisture, tree cores, sap flow, LAI, biomass, and meteorological data).
- Estimated biophysical variables (LAI, radiation interception, vegetation cover) from remote sensing data and calibrated against in-situ measurements.
- Computed Time Integrated NDVI (TIN) fitting curves to describe different ecosystems, including *Acacia caven* shrublands, *Pinus radiata* plantations, and *Nothofagus glauca* native forests.
- Managed the project database (GIS data, time series, and spreadsheets).
- Calibrated and installed scientific instrumentation (rain gauges, soil moisture probes, sap flow sensors).
- Developed scripts to pre-process, post-process, visualize, and analyze retrieved data.

Chile Oliva **Santiago, Chile**

Intern *2019*

"Olive oil yield prediction model... to increase competitiveness and efficiency."

- Measured olive tree yield predictor variables in the field, such as tree height, branch length, fruit count, and tree volume.
- Wrote R scripts to visualize meteorological data from stations near the studied orchards.

Publications

2024: **Rivero Iribarne, Lucas.** Improving WOFOST biomass and yield simulations: Assimilation leaf area index and canopy nitrogen with the ensemble Kalman filter. [*MSc. Thesis*]

2023: Ramirez, E., Galleguillos, M., Seguel, O., Faúndez, C., & **Rivero Iribarne, Lucas**. Dissecting the water balance of pine plantations and thorny forest in a Mediterranean landscape. *AGU Fall Meeting 2022, Chicago, IL.*

2021: Fassnacht, F. E., Poblete-Olivares, J., **Rivero, L.**, Lopatin, J., Ceballos-Comisso, A., & Galleguillos, M. Using Sentinel-2 and canopy height models to derive a landscape-level biomass map covering multiple vegetation types. *International Journal of Applied Earth Observation and Geoinformation, 94, 102236.*

2020: **Rivero Iribarne, Lucas.** Mapping of crop coefficient (Kc) from remote sensing in avocado in Mediterranean Chile. [Spanish] [*BSc. Thesis*] *Repositorio Académico - Universidad de Chile.*

Education

Wageningen University and Research (WUR) — **Wageningen, Netherlands**
Master in Geo-information Science — *2022 – 2024*
Thesis: Improving WOFOST biomass and yield simulations: Assimilation leaf area index and canopy nitrogen with the ensemble Kalman filter.
- **Relevant Courses:** Ecohydrology, Big Data, Geoscripting, Machine Learning, Deep Learning, Advanced Earth Observation.

OpenGeoHub — **Poznan, Poland**
Summer School: Processing and visualizing large geospatial data — *Aug 2023*
- Attended workshops on spatial data processing in Python and R, raster/vector data cubes, and geoprocessing in GRASS GIS.
- **Achievement:** 2nd Place in the Hackathon.

Centre d'Etudes Spatiales de la Biosphère (CESBIO) — **Online**
Summer School (Online) — *July 2021*
Topic: Remote sensing observations for the monitoring of water and carbon cycles over eco-agro-systems.
- Practical sessions on drought monitoring and evapotranspiration retrieval from space.

Pontificia Universidad Católica de Valparaíso — **Online**
Web Mapping Course — *Dec 2020*
Spatial analysis and web-mapping introduction with Google Earth Engine (GEE) and R Shiny.
- **R:** Developed an R Shiny app to visualize recycling points in Valparaíso.
- **GEE:** Performed time series analysis and supervised/unsupervised classifications.

Universidad de Chile — **Santiago, Chile**
Agricultural Engineering — *2014 – 2020*
Thesis: Mapping of crop coefficient (Kc) from remote sensing information in avocado in Mediterranean Chile.
- **Relevant Courses:** Edaphology, Agroclimatology, Soil Physics, Ecosystem Ecology, Remote Sensing, Advanced GIS, Precision Farming.
- **Teaching Assistant:** Irrigation and Drainage; Precision Farming.

Skills

Languages: Spanish (Native), English (Proficient), French (Slowly getting there)

Programming: Python, R, SQL

GIS & Remote Sensing: QGIS, ArcGIS, GRASS GIS, Google Earth Engine, SNAP

Crop Growth Models: WOFOST, SWAP, SWAT+

Cloud Computing: GCP, AWS (Basic)

Tools & Software: Docker, dbt, Airflow, HYDRUS 1D, CloudCompare, LaTeX, Excel